

Australian dollar drivers: initial model and estimation notes

ESSA Short Supply article — initial estimation

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1 Purpose and scope

These notes document the initial empirical model used for the ESSA *Short Supply* article on the Australian dollar.

The aim is not to build a definitive structural model of the Australian exchange rate. The aim is to use a compact small-open-economy system to ask two questions:

1. Which types of shocks tend to account for unexpected movements in the Australian real trade-weighted index (RTWI) at different forecast horizons?
2. Which shocks contributed to major historical Australian-dollar episodes?

The analysis builds on ideas used in Manalo, Perera and Rees (2015), while changing the variables, sample and substantive focus. The earlier replication work is stored separately at:

https://github.com/branesh97/replication_manalo_perera_rees_2015

The code in this folder should therefore be viewed as **initial article-stage estimation**. A later journal-oriented project may expand the information set and revisit identification.

2 Data and sample

The estimation sample is **1995Q1–2026Q2**, giving 126 quarterly observations.

The baseline system contains seven variables:

	Position	Variable	Role
	1	US real GDP gap	foreign activity
	2	RBA commodity price index	commodity-price conditions
	3	VIX	global financial risk
	4	Australian real GDP gap	domestic activity
	5	Australian trimmed-mean inflation	domestic inflation
	6	AU–US two-year government-bond yield spread	relative financial conditions
	7	Australian real TWI	exchange rate

The first three variables form the **foreign block**. The last four form the **domestic block**.

2.1 Transformations

The baseline transformations are:

- US real GDP: $100 \times \log(\text{real GDP})$, then detrended using a quadratic time trend;
- commodity prices: $100 \times \log(\text{index})$;
- VIX: $100 \times \log(\text{VIX})$;
- Australian real GDP: $100 \times \log(\text{real GDP})$, then detrended using a quadratic time trend;
- trimmed-mean inflation: observed quarterly series used in the dataset;
- AU–US two-year yield spread: level, in percentage points;
- real TWI: $100 \times \log(\text{RTWI})$.

For real GDP, the estimated gap is the residual from

$$100 \log(Y_t) = \alpha + \beta_1 t + \beta_2 t^2 + \tilde{Y}_t,$$

where $\tilde{Y}_t$ is the GDP-gap measure used in the VAR.

The exact raw series and source notes are stored in `data/data_collected.xlsx`.

3 Reduced-form VAR

Let

$$y_t = \begin{bmatrix} f_t \\ d_t \end{bmatrix},$$

where f_t is the foreign block and d_t is the domestic block.

The baseline reduced-form model is a VAR with two lags:

$$y_t = c + A_1 y_{t-1} + A_2 y_{t-2} + u_t,$$

with

$$E(u_t) = 0, \quad E(u_t u_t') = \Sigma_u.$$

The lag order is fixed at

$$p = 2.$$

This is consistent with the specification used in the initial structural analysis and is checked against one- and three-lag alternatives in the robustness exercises.

4 Small-open-economy restriction

Australia is treated as a small open economy. Domestic Australian variables are not allowed to affect the foreign block with a lag.

Partitioning each lag matrix gives

$$A_\ell = \begin{bmatrix} A_{ff,\ell} & 0 \\ A_{df,\ell} & A_{dd,\ell} \end{bmatrix}, \quad \ell = 1, 2.$$

Therefore:

- foreign equations contain a constant and lags of the three foreign variables only;
- domestic equations contain a constant and lags of all seven variables.

The restricted coefficients are stored as exact zeros.

This block-exogeneity restriction is imposed in both `olsvar_soe.m` and `bvar_soe.m`, and it is also preserved when bootstrap samples or posterior draws are generated.

5 Structural identification

The reduced-form innovations are mapped into orthogonal structural shocks as

$$u_t = B\varepsilon_t,$$

where

$$E(\varepsilon_t \varepsilon_t') = I$$

and

$$BB' = \Sigma_u.$$

The baseline model uses a recursive Cholesky identification with the ordering

[US GDP, commodity prices, VIX, Australian GDP, inflation, AU–US 2yr spread, RTWI].

Because B is lower triangular, variables earlier in the ordering can affect variables later in the ordering contemporaneously, while variables later in the ordering affect earlier variables only with a lag.

The baseline interpretation is therefore:

- US activity can affect all other variables contemporaneously;
- commodity prices can respond contemporaneously to US activity and can affect later variables;
- VIX can respond contemporaneously to preceding foreign shocks and can affect the domestic block;
- Australian activity is ordered before inflation and financial variables;
- trimmed-mean inflation is ordered before the yield spread;
- the AU–US two-year spread can affect the RTWI contemporaneously;
- the RTWI is ordered last.

The final structural shock is therefore an **exchange-rate-specific shock**: a contemporaneous RTWI innovation that is not attributed to the preceding macroeconomic shocks in this recursive system.

It should not be interpreted as a single primitive economic force.

6 Ordering robustness

The recursive ordering is an identifying assumption, so the main relationships are checked under four orderings:

1. commodity prices before VIX; spread before RTWI;
2. VIX before commodity prices; spread before RTWI;
3. commodity prices before VIX; RTWI before spread;
4. VIX before commodity prices; RTWI before spread.

The historical narrative is not based solely on one arbitrary contemporaneous ordering. However, the exact allocation between closely related contemporaneous shocks can change, especially for the relative-yield and exchange-rate-specific components.

7 Impulse responses

The moving-average representation is

$$y_t = \mu + \sum_{h=0}^{\infty} \Psi_h B \varepsilon_{t-h}.$$

The impulse response of variable j to structural shock k at horizon h is the corresponding element of

$$\Psi_h B.$$

For interpretation, the main IRF checks normalize shocks to economically intuitive magnitudes:

- commodity-price shock: **+10 log points**;
- VIX shock: **+10 log points**;
- AU-US two-year spread shock: **+1 percentage point = +100 basis points**.

The IRFs are used primarily as consistency checks. The article itself focuses more heavily on FEVDs and historical decompositions.

8 Forecast-error variance decomposition

For forecast horizon H , the structural forecast error for the RTWI can be written as the sum of the effects of orthogonal structural shocks over the horizon.

For RTWI variable j and shock k , the FEVD share is

$$\theta_{j,k}(H) = \frac{\sum_{h=0}^{H-1} (e'_j \Psi_h b_k)^2}{\sum_{m=1}^N \sum_{h=0}^{H-1} (e'_j \Psi_h b_m)^2},$$

where b_k is column k of B and e_j selects the RTWI equation.

The FEVD therefore answers:

What share of unexpected RTWI movements at horizon H is associated with each structural shock?

The article plots horizons from one to 20 quarters.

The principal displayed shocks are:

- commodity prices;
- VIX / global risk;
- relative yields;
- other exchange-rate shocks.

The shares across those four panels do not sum to 100 per cent because the full model also contains US activity, Australian activity and inflation shocks.

9 Classical SVAR uncertainty

The classical SVAR uses a two-stage residual bootstrap adapted to the SOE restriction.

9.1 First stage: finite-sample bias correction

Bootstrap samples are generated recursively by resampling entire reduced-form residual vectors. Resampling residuals by row preserves their contemporaneous covariance.

The SOE-restricted VAR is re-estimated for each artificial sample.

If $\widehat{\Phi}$ denotes the original VAR coefficients and $\widehat{\Phi}^{*(b)}$ denotes a first-stage bootstrap estimate, the estimated bias is

$$\widehat{\text{bias}} = \widehat{\Phi} - \frac{1}{B_1} \sum_{b=1}^{B_1} \widehat{\Phi}^{*(b)}.$$

The bias-corrected coefficient matrix is

$$\widehat{\Phi}_{BC} = \widehat{\Phi} + \widehat{\text{bias}}.$$

If the full correction produces an unstable VAR, the correction is shrunk towards the original estimate until stability is restored.

9.2 Second stage

Artificial samples are then generated from the bias-corrected dynamics and the restricted VAR is re-estimated in each replication.

The article-oriented bootstrap uses central:

- 70% intervals;
- 95% intervals.

The 70% interval is shown as a descriptive uncertainty region. It is **not** described as a conventional statistical-significance criterion. Conventional significance discussion is based on whether the 95% interval excludes zero.

10 Historical decomposition

For a fitted structural VAR, the observed series can be decomposed into:

1. deterministic/baseline dynamics;
2. the accumulated contribution of each structural shock.

For the RTWI,

$$RTWI_t = RTWI_t^{baseline} + \sum_{k=1}^N HD_{k,t}.$$

The baseline contains the effect of the constant and initial conditions when subsequent structural innovations are set to zero.

The article groups the structural contributions as:

- global activity;
- commodity prices;
- global risk (VIX);
- domestic macro;
- relative yields;
- other exchange-rate shocks.

“Domestic macro” combines the Australian GDP and inflation shock contributions.

For an episode from t_0 to t_1 , the contribution of shock k to the episode change is

$$\Delta HD_k = HD_{k,t_1} - HD_{k,t_0}.$$

The corresponding observed change is

$$\Delta RTWI = RTWI_{t_1} - RTWI_{t_0}.$$

The sum of the baseline change and all structural contributions reconstructs the observed episode change up to numerical precision.

11 Bayesian VAR

The Bayesian specification uses the same:

- seven variables;
- sample;
- lag order;
- SOE restrictions;
- recursive structural ordering.

The Bayesian component is introduced to shrink the relatively large number of VAR coefficients and to propagate parameter uncertainty into the FEVD and historical-decomposition exercises.

12 BVAR prior

The coefficient prior is Minnesota-style.

For an allowed coefficient on variable j at lag ℓ in equation i , the prior variance is

$$\text{Var}(\beta_{ij,\ell}) = \frac{\lambda^2}{\ell^{2d}} \frac{\sigma_i^2}{\sigma_j^2},$$

where:

- λ is the overall tightness parameter;
- d is the lag-decay exponent;
- σ_i^2 and σ_j^2 are residual variances from univariate AR models.

The baseline values are

$$\lambda = 0.20, \quad d = 1.$$

The intercept prior is diffuse, with prior variance

$$10^6.$$

12.1 Prior means

The own-first-lag prior means are set explicitly as

$$[0, 1, 0, 0, 0, 0, 1]'$$

Thus:

- commodity prices have own-first-lag prior mean 1;
- the RTWI has own-first-lag prior mean 1;
- US GDP gap, VIX, Australian GDP gap, inflation and the yield spread have own-first-lag prior mean 0;
- all other lag coefficients have prior mean 0.

This choice avoids imposing random-walk prior means on detrended GDP gaps, the VIX, inflation or the spread.

12.2 Prior for the covariance matrix

The reduced-form covariance matrix has an inverse-Wishart prior.

The degrees of freedom are

$$\nu_0 = N + 2.$$

The scale matrix is chosen so that the prior mean diagonal is approximately the set of univariate AR residual variances.

13 BVAR posterior simulation

The restricted BVAR is sampled with a Gibbs algorithm alternating between:

1. coefficients conditional on the covariance matrix;
2. the covariance matrix conditional on coefficients.

The final article baseline uses:

Setting	Value
Lag order	2
Minnesota tightness λ	0.20
Lag decay	1
Burn-in	5,000
Retained posterior draws	10,000
Thinning	1
Stability conditioning	yes
Maximum accepted companion root	0.9999
Residual draws stored	yes

The final posterior is conditioned on stability. Coefficient draws with a maximum companion root at or above 0.9999 are rejected and redrawn.

In the baseline run, approximately one-fifth of attempted coefficient draws were rejected by the stability condition. The exact rejection rate is an output diagnostic rather than a fixed modelling assumption.

14 BVAR FEVD uncertainty

For every retained posterior draw:

1. obtain the draw of the VAR coefficients and covariance matrix;
2. compute the Cholesky impact matrix;
3. calculate the RTWI FEVD;
4. store the FEVD share for each shock and horizon.

The article reports:

- posterior median;
- central 70% credible interval;
- central 95% credible interval.

The classical SVAR point estimate is shown as a dashed comparator in the article FEVD figure.

15 BVAR historical decomposition

Historical decomposition is also computed over posterior draws.

For the **stacked article episode chart**, the decomposition uses the **posterior-mean parameter model**:

$$\bar{\Phi} = E(\Phi \mid Y), \quad \bar{\Sigma} = E(\Sigma \mid Y).$$

The reason is accounting: a single coherent parameter model produces structural contributions that sum exactly to the observed historical decomposition.

Posterior medians of individual historical contributions generally do not obey an exact adding-up identity when stacked together, because the median of a sum need not equal the sum of medians.

Posterior uncertainty is therefore reported separately using posterior-draw distributions and credible intervals.

16 Main historical episodes

The article version uses a compact set of episodes, including:

- GFC;
- commodity-boom appreciation;
- post-boom depreciation;
- COVID shock;
- COVID rebound;
- recent period.

A separate diagnostic splits the recent period around 2025Q3 to check whether the large exchange-rate-specific contribution is simply an artefact of aggregating 2022–26. The qualitative result survives that split: relative yields contribute positively to the latest appreciation, but a sizeable “other exchange-rate shock” remains.

This diagnostic is useful for interpretation but need not appear as a separate bar in the magazine article.

17 Robustness exercises

The initial analysis includes several robustness checks.

17.1 Lag length

The classical model is re-estimated with:

$$p \in \{1, 2, 3\}.$$

The precise FEVD shares and IRF paths vary, but the main qualitative interpretation is not driven solely by the baseline two-lag choice.

17.2 Recursive ordering

Commodity/VIX ordering and spread/RTWI ordering are changed.

This is important because recursive structural labels depend on contemporaneous ordering.

17.3 Classical bootstrap stability

The distribution of companion roots is examined, and intervals using only stable bootstrap draws are compared with intervals using all draws.

This diagnostic indicated that wide uncertainty bands were not driven solely by a small set of explosive bootstrap draws.

17.4 BVAR prior tightness

The substantive FEVD and historical-decomposition results are checked for

$$\lambda \in \{0.10, 0.20, 0.30\}.$$

The central historical narrative remains similar across these values.

17.5 SVAR versus BVAR

The BVAR is not used to replace the classical SVAR mechanically. It is used as a shrinkage-based robustness specification.

The article emphasises results that are broadly consistent across the two frameworks.

18 Interpretation of the exchange-rate-specific shock

The final structural RTWI shock is labelled **other exchange-rate shocks** in the article.

This label is deliberately broad.

It captures RTWI innovations not contemporaneously attributed to the model's:

- US activity shock;
- commodity-price shock;
- VIX shock;
- Australian activity shock;
- inflation shock;
- AU–US two-year spread shock.

Possible economic mechanisms inside this residual include, among other things:

- broader US-dollar movements;
- foreign interest-rate developments not captured by the bilateral AU–US spread;

- China- or renminbi-related developments;
- portfolio rebalancing;
- currency hedging;
- carry trades;
- expectations that move financial markets before quarterly macroeconomic variables adjust.

The present model **cannot distinguish these mechanisms**.

This is especially important when interpreting the latest appreciation, for which the exchange-rate-specific component remains large.

19 What the initial model is designed to establish

The initial ESSA analysis is designed to support a relatively modest claim:

The forces associated with Australian-dollar movements are horizon- and episode-specific.

The empirical work is intended to show that:

- exchange-rate-specific and risk shocks matter relatively more at short horizons;
- commodity-price shocks become increasingly important over longer horizons;
- historical episodes have very different decompositions;
- no single driver provides a complete description of the Australian dollar over the full sample;
- the latest appreciation contains a sizeable component not explained by the conventional measured drivers in this parsimonious system.

20 Limitations and possible journal-paper extensions

A future journal-oriented model could consider:

- major-trading-partner activity rather than US GDP alone;
- China activity and renminbi variables;
- effective foreign interest rates or multi-country yield differentials;
- a broad US-dollar factor;
- capital-flow and portfolio-positioning measures;
- currency-hedging variables;
- alternative global-risk measures;
- sign restrictions, narrative restrictions or external instruments;
- time variation or regime dependence;
- a larger BVAR with a richer foreign block.

These extensions are intentionally left outside the initial ESSA article analysis.

21 Key files

The main empirical files are:

- `svar_initial.m` — baseline classical SOE-SVAR;
- `svar_bootstrap.m` — bootstrap uncertainty;
- `svar_ordering_robustness.m` — recursive-ordering sensitivity;
- `svar_lag_robustness.m` — lag sensitivity;
- `svar_hd.m` and `svar_hd_final.m` — classical historical decomposition;
- `bvar_baseline.m` — final stable SOE-BVAR;
- `bvar_irf_check.m` — BVAR/SVAR IRF consistency;
- `bvar_fevd_hd.m` — posterior FEVD and historical decomposition;
- `bvar_fevd_hd_articlefig.m` — article FEVD and selected-episode HD figures;
- `bvar_lambda_subs_robustness.m` — prior-tightness robustness.

22 References

Manalo, J., Perera, D. and Rees, D. M. (2015), “Exchange Rate Movements and the Australian Economy”, *Economic Modelling*, 47, 53–62.

Kilian, L. (1998), “Small-Sample Confidence Intervals for Impulse Response Functions”, *Review of Economics and Statistics*, 80, 218–230.

The earlier replication repository is available at:

https://github.com/branes97/replication_manalo_perera_rees_2015